

ORIGINAL RESEARCH

An on-chain trading model of real world asset backed digital assets

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Abstract

Metaverse is a digital value interaction network based on blockchain technology, with an important economic system component. While both traditional financial industries and crypto-native industries have made significant progress by leveraging blockchain, the value stream of each remains limited to separate ecosystems. To bridge this gap between off-chain and on-chain economic systems, an on-chain trading model was proposed using HD key derivation technique for direct uploading onto chains without going through centralized services for IoT data transmission. To improve the current status of NFTs as static assets, a token protocol binding each NFT with a unique account address was proposed. Additionally, oracle technique was leveraged with a decentralized and distributed trust model spanning across on-chain and off-chain components which securely pushes data between smart contracts and Web-APIs. A decentralized trading model was developed based on smart contracts implementing automated market makers according to CFMM algorithm. Parallel transaction computing was executed based on the DAG model to ensure high operational performance and security standards of underlying blockchain. Finally, the on-chain trading system of real world asset backed digital assets was developed integrating all the above key techniques that correspond to crucial functions of a complete economic system in Metaverse.

1 | INTRODUCTION

The Working Committee of Metaverse under China Electronics Standardization Institute has clarified the essence of Metaverse from an ontological perspective in its report “Linking Metaverse: Research and Exploration” [1]. The report points out that Metaverse is an integrated logical expression of digital ecology, which is a trusted digital value interaction network with blockchain as its low layer technology. The core elements of Metaverse are organization, identity, assets and activities, and the key directions are technology, content, economy, collaboration and governance systems. Among them, a sustainable economic system for Metaverse should include basic functions such as digital value creation, digital asset ownership confirmation, digital asset trading and circulation.

Both traditional industries and crypto-native industries have made significant progress on the economic society using blockchain. Blockchain technology is playing an increasingly important role in the financial industry. Javaid et al. [2] summarized several featured services, such as cross-border financial

transactions, trade finance platforms, proper reporting of credits, clearing and settlements, and digital identity verification. Digital financing, especially supply chain financing, has been a mature business model for blockchain adoption, out of various development and advancement made by blockchain practices. Alao et al. [3] described a novel decentralized autonomous organization, a blockchain-agnostic special purpose vehicle underpinned by a trio of autonomous mechanisms, including mobilization, collateralization and settlement. These mechanisms enable seamless finance mobilization for the special purpose vehicle from a location-independent crowd, revenue collection from the electricity buyer in a risk-mitigated manner, and disbursement of eventual project revenues to investors. Liu et al. [4] considered a three-level supply chain composed of a manufacturer, a distributor, and a capital-constrained retailer, and explores the operational strategies of blockchain platform finance (BPF), a supply chain finance mode enabled by blockchain technology. This study provided a theoretical basis for decision makers to implement blockchain supply chain finance at three levels: joint financing and operation

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decision-making, financing mode selection, and financing efficiency improvement. Shibuya et al. [5] provided conditions for the remote-tier financing to be favoured by the supply chain, compare preferences of individual firms and the supply chain, and discuss how this affects Blockchain adoption for supply chain financing. The result suggested that even if technology advancement allowed a single downstream firm to finance the entire supply chain, economic forces impose a limit on the number of tiers in a supply chain financing solution. Choi [6] developed analytical models to analyse impacts of blockchain on supply chains selling fashionable products. The study proved that the blockchain-supported supply chain incurs a lower level of operational risk than the traditional supply chain. The study also showed that if the service fees by banks are sufficiently high, adopting blockchain technology is a mean-risk dominating policy which brings a higher expected profit and a lower risk for the supply chain and its members. Rao et al. [7] proposed a blockchain supply chain financing risk assessment and behaviour prediction algorithm. By analysing the development status of supply chain business, it puts forward corresponding preventive measures to effectively improve the efficiency of supply chain financing. Alao et al. [8] proposed a novel decentralized finance instrument, a blockchain special purpose vehicle that streamlines the processes in the financial layer of a traditional special purpose vehicle-finance mobilization, revenue collection, and revenue disbursal. Specifically, the proposed decentralized finance instrument facilitates the mobilization of finance for the special purpose vehicle from a location-independent crowd, revenue collection from the electricity off taker in a risk-mitigated manner, and disbursal of eventual project revenues to investors. Fu et al. [9] proposed a blockchain-based financing scheme (BFS) for logistics company, utilizing efficient and interpretative smart contract technology, improved privacy information query and invocation algorithms. The scheme could significantly simplify the financing procedures and lower the thresholds to financing for logistics companies, while protecting the privacy of crucial data. Simulation tests showed that BFS could run supply chain blocks at a stable transaction throughput of around 280 RPS, which met the financing needs of logistics companies.

The launch of the Turing complete Ethereum blockchain in 2015 extended the scope of blockchain-based financial systems beyond cryptocurrencies [10]. Decentralized finance, more commonly called DeFi, provides financial services on a blockchain settlement layer without centralized intermediaries or institutions and generally encoded in open source software protocols and smart contracts. Since the DeFi summer, DeFi has been developing fast in six major categories [11], that is, stablecoin, exchange, credit, derivative, insurance, and asset management. Auxiliary services such as wallet and oracles are also important DeFi tracks. The DAI stablecoin [12], which is issued by MakerDao, is pegged to the US dollar. DAI, a collateralized stablecoin, is based on a set of smart contracts on the Ethereum blockchain. Everyone can generate new DAI tokens on condition that a user shall send Ether to one of the smart contracts. These funds serve as collateral for a loan in DAI that the user can get with a relatively low interest rate as sta-

bility fee. Despite blockchain based digital assets trading since 2009, there has been a functional gap between on-chain transactions and trust based centralized exchanges. Uniswap [13] is a decentralized exchange providing automated market maker service. Uniswap based on user provided liquidity reserves enables the trading of blockchain token without relying on market makers, bids or asks. A pricing mechanism of Constant Function Market Maker(CFMM) is leveraged to manage the liquidity reserves. To request a loan in DeFi platforms, the user is usually required to deposit overcollateralized assets. Flash Loan [14] is developed to enable a non-collateral borrowing service. Additionally, a considerable amount of assets can be lent to users by Flash Loan as long as the borrowed assets can be paid back within the current transaction. Otherwise, the platform will instantly revert the transaction to get the lent assets back. Synthetix [15] allows users creating and exchanging synthetic versions of assets whose price can track currencies, cryptocurrencies and commodities. Synthetix involves two distinct types of token. Users first purchase and lock up Synthetix's native token SNX into the Synthetix contract that gets staked as collateral to back other Synths tokens. These Synths are synthetic assets created through Synthetix platform. The price of Synths is determined through oracles that report external real-world price of asset. Nexus Mutual [16] is creating decentralized insurance on Ethereum by using a risk-sharing pool. The pool is governed by its members where membership rights are represented by the NXM token. The mutual is initially launching with smart contract cover, allowing anyone to purchase insurance on any public Ethereum smart contract. DeFi users can then get protection on their funds being lent out. Set Protocol [17] is mainly designed for semi-automated strategies with deterministic portfolio rebalancing triggered by predefined threshold values and time locks. However, the protocol is also used for active management. Betoken operates as a single fund of funds managed by a community of asset managers through a meritocratic system. The more successful an individual fund manager is, the greater their future influence on allocating the collective resources.

Through the investigation, Blockchain has been applied in traditional financial industry as emerging technology, and the majority published financial application cases focus on international payment, time and expense reduction, data traceability, privacy maintenance etc. Though the trusted information supported by blockchain can be a strong evidence for financing credit, the actual value stream still flows as real world assets off-chain and within separate consortium ecosystem of specific business scenarios. On the other hand, large amounts of cryptocurrencies are actively being swapped, lent, borrowed, collateralized for staking or liquidity provision etc., worldwide. For most cases, cryptocurrency is created as native token of certain public chain ecosystem and traded for financial purposes within. Without being linked to any real world assets, cryptocurrency has not demonstrated its potential value proposition in traditional industries. Therefore, the digital finance mode of traditional financial industry and the cryptocurrency mode of public chain ecosystem are operating independently with separate advantages. Here, an on-chain solution for the trading

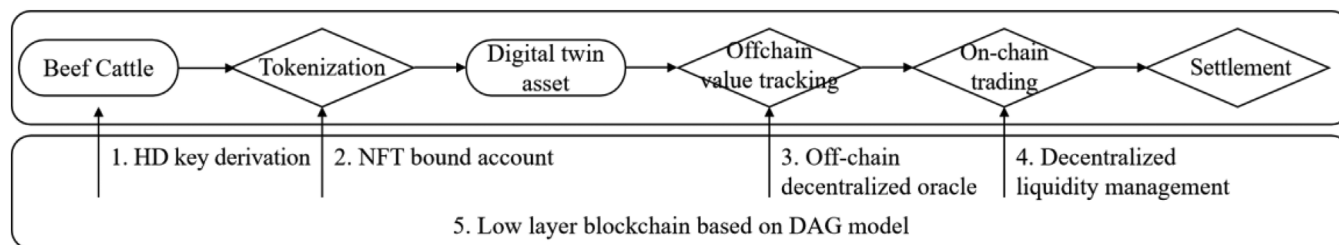


FIGURE 1 Technical diagram of digital twin asset on-chain trading system.

of digital assets that are backed by real world assets is proposed to bridge the gap between off-chain economic system and on-chain economic system. Section 2 introduces an overall technical scheme of the solution. Section 3 further illustrates key techniques utilized in the scheme. Section 4 describes an on-chain trading system designed for the implementation of the solution. Section 5 finally demonstrates main conclusions of the study.

2 | TECHNICAL SCHEME

Real world asset (RWA), in essence, entails the tokenization of real-world off-chain assets on the blockchain. The scope of RWA encompasses a diverse array of asset types, with a primary focus on fixed-income assets such as government and corporate bonds. Additionally, RWA extends to assets that necessitate institutional packaging, including biological assets, collateral assets, and real estate rents. Moreover, RWA includes the tokenization of volatile assets directly, such as houses, carbon credits, and stocks. Beef cattle have emerged as valuable financial assets in real world, with ranches commonly resorting to beef cattle financing as a means to overcome capital shortages. This practice involves securing funding by collateralizing beef cattle and seeking loans from financial institutions. However, due to the lack of robust regulatory mechanisms for monitoring collateralized assets, financial institutions struggle to effectively control risks associated with the beef cattle farming industry, thus hindering the provision of high-quality financial services. In practical implementation, beef cattle financing encounters significant challenges pertaining to operational management, asset supervision, and mortgage loans. A digital financing scheme is new trend for the industry.

A technical solution was designed to enable on-chain transactions for real world assets, taking beef cattle as a representative of tangible asset, shown in Figure 1. The specific procedures and core techniques utilized are as follows: (1) The first step is to create an on-chain account for real world assets that can manage data, assets, security, and privacy protection. Hierarchical deterministic (HD) key derivation algorithm is applied. (2) Secondly, a digital twin of the real world asset should be created in the blockchain network so that it can participate in established financial activities such as financing or trading. The process of digital twin is realized through tokenization using a protocol of Non-fungible Token (NFT) bound account. (3) In order to maintain the value correlation between digital and

real world assets, market information from off-chain sources must be continuously collected using off-chain decentralized oracle technology, corresponding to the process of off-chain value tracking. (4) For on-chain trading of RWA backed digital assets, decentralized exchanges enhance liquidity for digital assets through automated market maker models while enriching financial services to expand business operations. (5) These core functions rely on smart contract technology where data communication has strict logical relationships, which requires high operational performance and security standards at the underlying blockchain level, based on Directed Acyclic Graph (DAG) model.

3 | KEY TECHNIQUES

3.1 | Hierarchical deterministic key derivation

The traditional biological data monitoring of beef cattle is uploaded directly to the Internet of Things (IoT) gateway through collection equipment and then interacted with blockchain after being summarized. The farm owner has a unique private key on the IoT gateway to manage the collection of beef cattle data on the blockchain, enabling management of multiple beef cattle assets. However, since the IOT gateway is usually a traditional centralized service, there is a risk that critical data may be tampered with. On the other hand, if each beef cattle is configured with an independent private key and directly uploaded to the chain through an IoT device, it can ensure the authenticity of chain-uploaded data. However, in cases where there are many head of cattle involved, maintaining these private keys would incur high costs and could lead to situations such as loss or mismatching between private keys and individual animals.

HD key derivation technology improves upon deterministic wallets by using elliptic curve cryptography algorithms based on tree structures for key derivation. Through HD key derivation technology, one private key and path can be derived for each beef cattle's IoT device. This allows direct uploading onto chains without going through centralized services for IoT data transmission. The logic scheme is shown in Figure 2.

The HD Key Derivation Technology includes two parts: main key generation and sub-key generation; as shown in Figure 3.

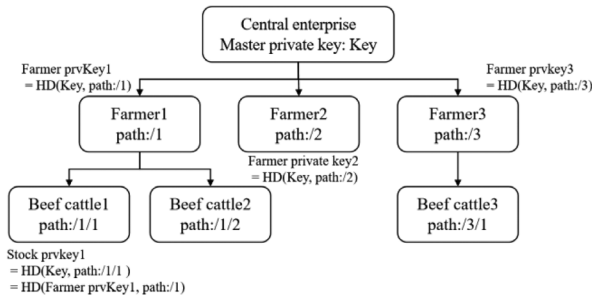


FIGURE 2 Key derivation in beef cattle scenario.

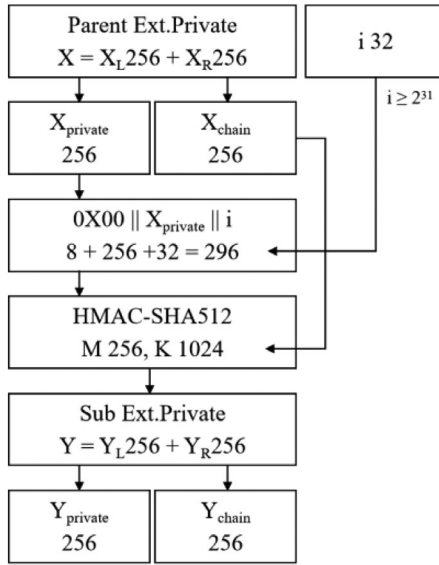


FIGURE 3 Logic scheme of HD key derivation [18].

The specific implementation path is as follows:

- (1) Mnemonic generation. Call the generateMnemonic method encapsulated in bip39.js. BIP39 defines how wallets create seed phrases and generate encryption keys [19].
- (2) Root seed generation. Call the mnemonicToSeedSync method encapsulated in bip39.js to generate a root seed from the mnemonic using the key extension function PBKDF2.
- (3) Master key generation. The root seed undergoes 2048 rounds of HASH calculation using HMAC-SHA512 algorithm, generating a 512-bit hash string. The left 256 bits are used as the master private key m , and the right 256 bits are used as the master chain code (Master Chain Code). The master public key M can be generated based on elliptic curve algorithm from the master private key m .
- (4) Subkey generation. Call the fromSeed method encapsulated in bip32.js, passing in either the master private or public keys as input for hierarchical derivation to obtain a 256-bit sub-private key and sub-chain code, along with corresponding sub-public keys, repeating this process multiple times. BIP32 lays out a framework for HD Wallets so that the wal-

let software can control multiple separate accounts using a single seed phrase [19].

- (5) Key path mapping. After generating tree-structured public-private key pairs information, call derivePath method in bip32.js for path mapping (following BIP44 protocol, defining the same organizational hierarchy for managing multiple accounts in deterministic wallets for all categories digital assets [19]) to obtain corresponding public-private keys for beef cattle assets.
- (6) Identity address generation. Calculate different cattle's identity addresses on blockchain network based on Keccak256 algorithm.

HD key derivation technology has several technical advantages including: (1) ensuring data authenticity on-chain and reducing management costs of private keys; (2) each cow's public key can be calculated through parent public keys and paths making it easy for third-party organizations to verify beef cattle information without leaking private keys; (3) HD key derivation technology supports multi-level derivation suitable for scenarios involving centralized enterprises, large farm owners, farmers etc., where extreme cases involve central enterprises or farm owners re-matching beef cattle asset private keys.

3.2 | Real world asset tokenization

Asset tokenization is the use of blockchain and smart contract technology to convert real-world assets into digital securities on the blockchain. Real-world assets include physical or non-physical assets. tokenization essentially records the digital ownership of an asset on an immutable blockchain. The current mainstream token protocols are ERC20, ERC721, and ERC1155 [20], which include key processes such as minting, issuance, trading, and management of digital assets that help improve their liquidity. ERC20 provides a standardized homogeneous token standard; ERC721 provides a non-fungible token standard; while ERC1155 provides a mixed-token protocol that allows for confirmation and control over any number of homogeneous or non-homogeneous tokens while supporting batch transfers and transactions.

Existing NFT protocols still only represent static assets in circulation and can only prove the current holder's ownership of specific NFT assets, limiting the application scenarios for NFTs. This section builds upon the ERC721 protocol by binding each NFT with a unique account address so that it can record value changes, sign transactions, hold on-chain assets, interact with other components or Dapps etc. When an NFT participates in a transaction all relevant parties can see all these records together with equivalent asset values and related information being transferred.

This proposal grants every ERC-721 token the full capabilities of a blockchain account by deploying unique, deterministically-addressed smart contract accounts for each ERC-721 token via a permissionless registry. Though, the constructor of each NFT may be sub private key from HD key derivation, the ownership can be transferred without

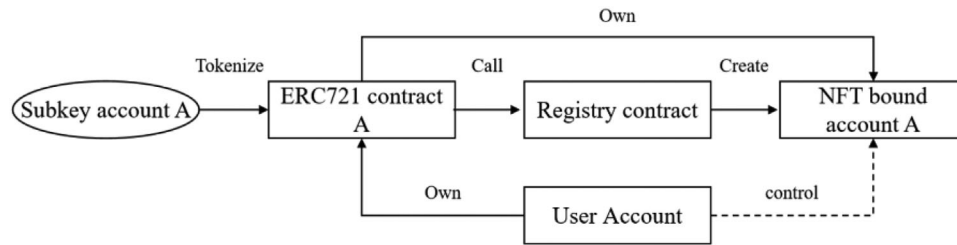


FIGURE 4 Logic scheme of digital twin asset tokenization.

hardcoding any provenance. Control of each token bound account is delegated to the owner of the ERC-721 token, allowing the owner to initiate on-chain actions on behalf of their token.

The registry smart contract serves as a single entry point for projects wishing to utilize token bound accounts, with the features of being permissionless, immutable, and having no owner. The contract has two functions: (1) create deploys a unique, deterministically-addressed smart contract account for an ERC-721 token. (2) Account—a read-only function that computes the token bound account address for an ERC-721 token.

The registry shall deploy all token bound account contracts using the create2 opcode so that the account address for every ERC-721 token is deterministic. The account address for each ERC-721 token shall be derived from the unique combination of token contract address, token ID, chain ID, and an optional salt, as shown in Equation (1).

Creation Code :

$$\text{code} = \text{abi.encodePacked}(\text{creationCode}, \text{abi.encode}(\text{salt}, \text{chainId}, \text{tokenContract}, \text{tokenId}, \text{implementation}))$$

Compute contract address from bytecode :

$$\text{_account} = \text{Create2.computeAddress}(\text{bytes32}(\text{salt}), \text{keccak256}(\text{code}))$$

Deploy contract

$$\text{_account} = \text{Create2.deploy}(0, \text{bytes32}(\text{salt}), \text{code})$$

(1)

The following diagram in Figure 4 illustrates the relationship between ERC-721 tokens, ERC-721 token owners, token bound accounts, and the Registry. For simplicity, a single token bound account is assigned to each NFT as an unambiguous identifier, though alternative architectures are adaptive for other scenarios.

3.3 | Off-chain decentralized oracles

To conduct a value assessment of biological assets mapped as digital assets in the metaverse, it is necessary to obtain real and reliable asset-related data, including both the biological feature data of the biological assets and the cost of breeding during their growth cycle, as well as market information. Normally, smart contracts deployed on blockchain lack the ability of connecting themselves with off-chain resources (like web) natively. Oracle technique aims to resolve this issue by acting as a decentralized

oracle network bridging on-chain smart contracts with the off-chain environment.

The oracle developed in the paper proposed a decentralized and distributed trust model spanning across on-chain and off-chain components which securely pushes the data between smart contracts and Web-API in order to create externally-aware tamper-proof smart contracts, as shown in Figure 5.

The oracle plays a crucial role in obtaining accurate and reliable off-chain information through a multi-level data aggregation process. This includes aggregating data from various sources, consolidating node operators' contributions, and leveraging the collective wisdom of the oracle network. The primary sources of original data consist of trusted platforms and recognized data providers, such as trading market websites, third-party authentication agencies' price feeds, and decentralized IoT systems. Ensuring the reliability and comprehensiveness of these data sources is essential for the accurate aggregation of original data.

Node operators form an integral part of the system, responsible for acquiring and disseminating external market data on the blockchain. They follow established rules to gather aggregated data from multiple independent channels. To enhance the quality of the aggregated data, node operators eliminate abnormal values and consider potential single-point failures before calculating the median value. This approach minimizes the impact of outliers and increases the overall accuracy of the final numerical value used for off-chain information.

The oracle network aggregation process further enhances the accuracy and reliability of the off-chain data. By setting a predefined threshold for the number of response nodes, the system ensures a sufficient level of participation from the oracle network. Once this threshold is met, the median value derived from the aggregated data becomes the final numerical value of the off-chain information. This consensus-based approach reduces the influence of individual nodes or potential malicious actors, enhancing the integrity and reliability of the data.

Additionally, measures are taken to optimize the use of on-chain computing resources. Off-chain information is only stored on the blockchain when there are significant updates beyond predefined deviation thresholds or when heartbeat thresholds are reached. This selective approach minimizes the burden on on-chain resources while still ensuring that important updates are recorded and accessible.

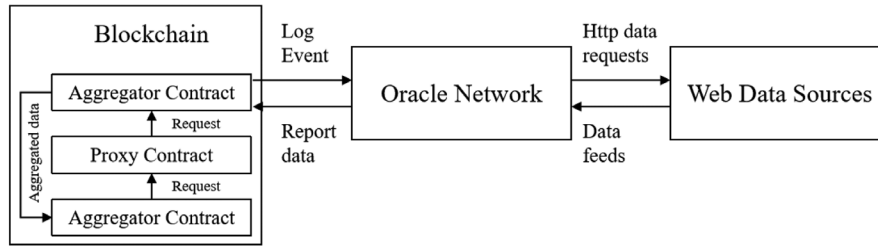


FIGURE 5 Logic scheme of off-chain decentralized oracle.

All nodes in the oracle network communicate using a peer to peer network. During the communication process, a lightweight consensus algorithm runs where each node reports its data observation and signs it. A single aggregate transaction is then transmitted, and a report in the transaction is signed by a quorum of oracle. The final report containing all oracles' observations is committed to the aggregator contract according to a randomized schedule. The aggregator verifies that a quorum of nodes signed the report and exposes the median value to consumers as an answer with a block timestamp and a round ID. The off-chain reporting mechanism preserves the trustlessness properties of oracle networks and saves a significant amount of network resources.

The oracle developed consisted of three contracts, that is, consumer contract, proxy contract and aggregator contract. The developed oracle consists of three contracts: the consumer contract, proxy contract, and aggregator contract. For each off-chain data or dataset, an Aggregator contract and a corresponding Proxy contract are deployed. Users do not directly use the aggregator contract but instead obtain data from the Aggregator through the Proxy contract, which helps isolate the complexity of the Aggregator. The Aggregator only provides a unified interface that returns the most recent price data from the oracle network.

3.4 | Decentralized trading model

An exchange is an important way to facilitate the trading and exchange of different assets in the metaverse. Traditional exchanges match trades through a centralized order book system. The order book records all buy and sell orders participating in the exchange, and both parties trade at a specific exchange rate while the exchange charges a service fee. However, if there is low activity on the exchange, users' orders may remain unfilled for long periods of time. To address this inefficiency with order book models, traditional trading markets have invented market maker models that essentially improve liquidity through third-party agents to enhance transaction efficiency. In order to ensure safe, transparent and publicly accessible market-making assets, this section proposes a decentralized trading model based on smart contracts that implement automated market makers.

The core of the decentralized trading model is creating digital asset liquidity which mainly includes factory contracts and liquidity pool contracts. A single liquidity pool should include two types of tradable token assets: Token A and Token B; using biological asset BAV as an example along with official digi-

tal currency DCEP issued by China's central bank where price would be represented as BAV/DCEP.

The factory contract is used to create flow pool contracts. Based on input parameters (BAV on-chain address " a ," DCEP on-chain address " b ," fee " f "), the factory contract uniquely determines asset flow pools by traversing all (a, b, f) information for each flow pool created so far; it then checks whether or not corresponding flow pools have been created yet for (a, b, f) . If no such flow pool exists yet for (a, b, f) , then using Keccak256 algorithm, it obtains hash value $\text{abi.encode}(a, b, f)$ which creates corresponding Pool contract addresses.

The liquidity pool contract is used to lock or release digital assets of liquidity providers, and calculates the overall liquidity according to CFMMs algorithm. Liquidity providers input market-making parameters into the liquidity pool contract, as shown in Figure 6a: BAV on-chain address a and intended added quantity $\text{amount}A$, DCEP on-chain address b and intended added quantity $\text{amount}B$, transaction fee f , owner's address of the liquidity asset provided, upper limit price P_u for providing liquidity asset and lower limit price P_l for providing liquidity asset. Tick values T_u and T_l corresponding to P_u and P_l are calculated according to Equations (2) and (3) to ensure consistency and discretization of trading prices in the liquidity pool. The position information of each liquidity provider is represented by (user address, T_u , T_l) key-value pair. When multiple users provide liquidity in the same price range, the Liquidity Pool Contract will merge storage. The presentation form of trading tick is shown in the Figure 6b.

$$P = \text{sqrt}(\text{BAV}/\text{DCEP}) \quad (2)$$

$$\text{Tick} = \log 1.0001P \quad (3)$$

After the digital assets are transferred to the liquidity pool contract from liquidity providers, ERC721 tokens are issued to the liquidity provider's account address as proof of their position in the liquidity pool and as evidence for receiving service fees. The NFT is burnt after all digital assets have been withdrawn by users.

3.5 | Parallel transaction based on DAG model

In the conventional process of "block packaging-network consensus-transaction storage," executing all transactions in the block is a prerequisite for on-chain of the block. The traditional

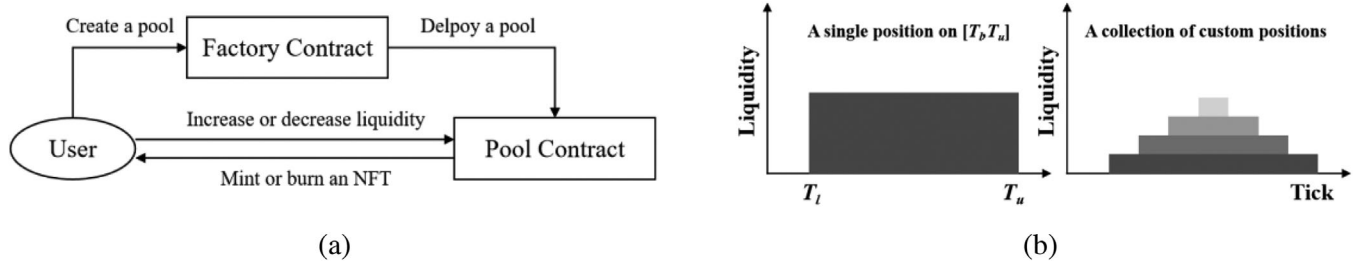


FIGURE 6 Logic scheme of liquidity management. (a) Procedures of providing liquidity. (b) Scheme of liquidity distribution.



FIGURE 7 Logic scheme of parallel transactions based on DAG.

transaction execution scheme is: the execution unit gets each transaction information from the block to be finalized one by one, and the state machine updates to the next state when all transactions are executed serially. Obviously, this kind of transaction execution method is not friendly to performance. Even if two transactions have no intersection, they can only be executed in order.

In practical applications, a transaction dependency graph is organized based on the mutually exclusive resources required for each transaction execution. In order to prevent circular dependencies in the graph, it can be stipulated that transactions involving the same mutually exclusive resources in the transaction list and sorted later must wait for earlier transactions to complete before being executed. This results in a directed acyclic graph reflecting transaction dependencies, namely Transaction DAG. Today's mainstream multi-core processor architecture has made full use of parallelization technology an effective solution to improve blockchain performance.

The core difficulty of parallel processing based on DAG model lies in how to determine dependencies between transactions. The complete technical solution is as follows, and the brief logic scheme is shown in Figure 7.

- (1) The block proposer simulates the execution of transactions in the packaged block to generate read and write sets.

Rewrite state-related operations and log-related operations of `*state.StateDB` for TxSimulator, record the operations in `ReadSet`, `WriteSet`, `BalanceSet`, and logs. Implement `core.ApplyTransactionForSimulator` method with logic consistent with `core.ApplyTransaction` but passing TxSimulator as a parameter without `statedb.Finalise(true)`. After completing transaction execution, all read and write operations on state and balance during transaction execution can be obtained through `ReadSet`, `WriteSet`, and `BalanceSet`.

- (2) Generate transaction DAG based on read and write sets.

The block proposal node initiates a block proposal and obtains a certain number of transactions from the transaction pool. By simulating the execution of transactions in parallel, the transactions are added to the stateDB's transaction cache in a serial manner. After simulating the execution of transactions, TxSimulator obtains the read-write set and balance changes of each transaction. It then determines whether there were any conflicting transactions applied to stateDB during parallel execution. If so, it abandons the simulated results of that transaction and puts it back into the transaction execution channel for re-execution later.

TxSimulator's transactions, as well as their corresponding read-write sets and balance changes, are applied to stateDB. The dependency relationship between these transactions is determined based on their respective read-write sets' dependencies. When applying TxSimulator's transactions to stateDB, its position index in stateDB's list is used as its actual version number; its read-write set information is updated in stateDB's cache; and its balance information is updated in stateDB's balance cache.

Finally, all completed transactions' DAG_Dependency are added to stateDB's DAG with an index equalling their actual version numbers. Once all this has been done, calling `getDag()` will return a direct dependency relationship DAG for all executed transactions.

- (3) Pack the DAG of transactions and block information (including the state root generated based on DAG) into a message for block broadcasting. Other nodes calculate transaction results in parallel based on DAG information. If the DAG information is empty or len equals 0, it means that the DAG function was not used to execute transactions within the block, so serial logic is used to execute transactions. Transactions with no dependent transactions (`dag_dependency` is empty) are executed concurrently first. After execution, dependency edges are reduced for completed transactions. If dependencies disappear, they are added to an execution channel for pending transactions

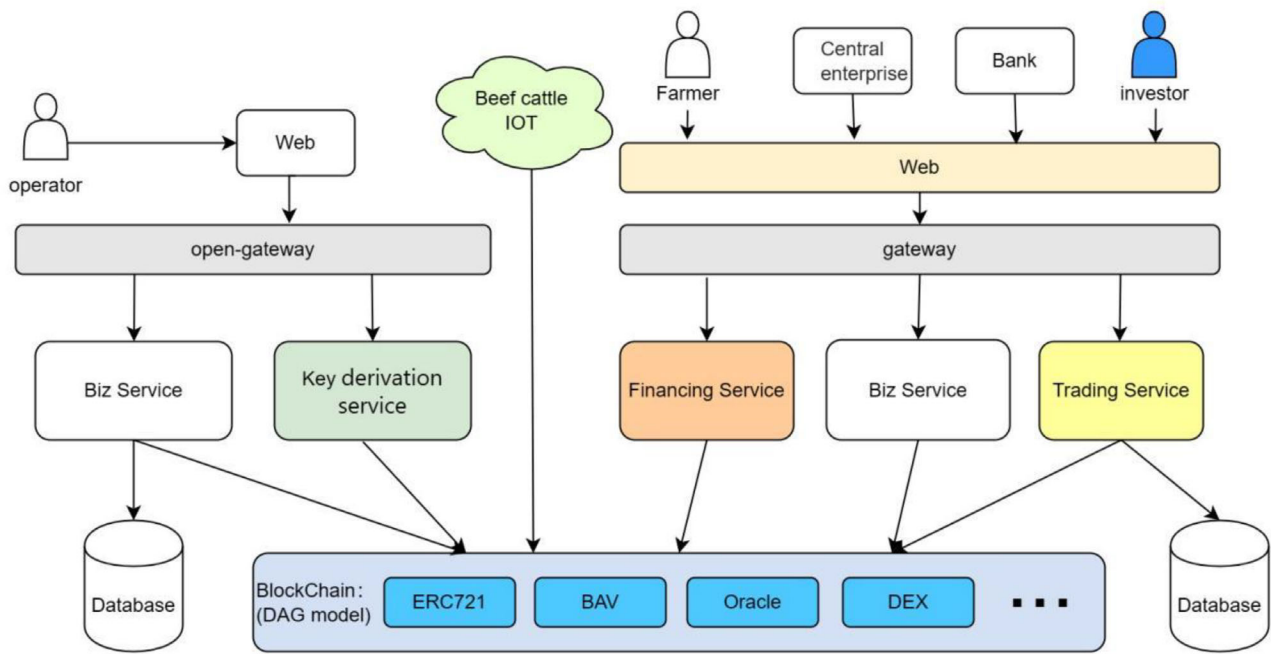


FIGURE 8 Logic architecture of on-chain trading system of real world asset backed digital assets.

until all transactions have been executed. The receipts generated by these transactions are stored according to their positions in the body for generating corresponding receipt hashes. A DAG structure is added to the block structure.

- (4) Each validator independently runs Tx based on DAG information and verifies whether the obtained state root matches that in the block.

4 | DESIGN OF ON-CHAIN TRADING SYSTEM

The on-chain RWA trading system combines several key components to ensure seamless and efficient operations. The first critical element is the HD key derivation service, which enhances security and enables robust key management for users. This ensures that sensitive information and transactions remain secure within the system. Moreover, the platform incorporates essential business services, such as financing and trading services, to facilitate smooth and transparent transactions. This enables users to engage in various financial activities and trading operations in a user-friendly and efficient manner. By integrating these services into the Metaverse ecosystem, users can conveniently access financial tools and opportunities while benefiting from the underlying blockchain's security and immutability. The platform's foundation relies on a low-layer blockchain that leverages a DAG model, enabling parallel transactions and increasing the network's scalability. This approach optimizes transaction speed and throughput, ensuring that the system can handle a significant volume of transactions with minimal delays. To interact with the blockchain and access its various functionalities, the system employs smart contracts,

which act as self-executing agreements with predefined conditions. This includes the implementation of ERC721, NFT bound accounts, oracle services, and decentralized exchanges, among others. These smart contracts provide a reliable and transparent framework for executing complex financial operations, asset management, and trading activities within the Metaverse. The logical architecture of this integrated platform is visually represented in Figure 8. By leveraging the power of blockchain technology and incorporating advanced features, the RWA on-chain trading system offers users a seamless, secure, and rewarding experience within the virtual realm.

Potential challenges in implementing IoT-based systems for monitoring real-world assets, such as beef cattle, revolve around the reliability and stability of collected metadata. IoT devices, often in the form of ear tags, enable remote monitoring of overall ranch operations and vital signs of cattle, facilitating remote supervision. However, the risk of ear tag damage or loss can lead to data loss or disruptions, including simultaneous detachment of multiple ear tags, which can impact the continuous tracking of beef cattle asset value. Overcoming the challenge of reliable data sources requires advancements in IoT technology and innovative operational models. For instance, IoT vendors have explored the option of implanting sensors into the stomach of beef cattle, ensuring a stable data collection environment, albeit at increased operational costs.

5 | CONCLUSION

Metaverse is a digital value interaction network that uses blockchain technology as its foundation. A sustainable economic system includes four main functions: digital value

creation, digital asset ownership confirmation, digital asset trading and circulation, which is a key direction of Metaverse. Both traditional financial industries and crypto-native industries have made significant progress by leveraging blockchain. However, the traditional financial sector focuses on data while actual value still flows through real world assets off-chain. Cryptocurrency, on the other hand, is created as native tokens of certain public chain ecosystems and traded for financial purposes within those ecosystems. To bridge the gap between off-chain and on-chain economic systems, an on-chain trading model was proposed for digital assets backed by real world assets.

In the model, Hierarchical Deterministic (HD) key derivation technique was used to enable direct uploading onto chains without going through centralized services for IoT data transmission using elliptic curve cryptography algorithms based on tree structures for key derivation. A token protocol was also proposed upon original ERC721 by binding each NFT with a unique account address so that it can record value changes, sign transactions, hold on-chain assets, interact with other components or Dapps etc. Oracle technique was leveraged with a decentralized and distributed trust model spanning across on-chain and off-chain components which securely pushes the data between smart contracts and Web-API in order to create externally-aware tamper-proof smart contracts. For active digital asset trading, a decentralized trading model is developed based on smart contracts that implement automated market makers according to CFMM algorithm. To ensure high operational performance and security standards of underlying blockchain, parallel transaction computing was executed based on the DAG model. Finally, the logical architecture of the on-chain trading system of real world asset backed digital assets integrates all key techniques mentioned above.

AUTHOR CONTRIBUTIONS

Dongsheng Hou contributed to the conception of the study, the design of the technical scheme, and wrote the manuscript. Wenjing Ma contributed to the HD key derivation technique and the technical implementation of the whole study. Wei Zhang contributed to the study of DAG model and low layer blockchain technique. Yixuan Li offered significant industrial experience and support for the study. Yu Du contributed to the conception of the study and offered practical advice on the adoption of the study. Yukun Hao contributed to the development and implementation of the system based on the study.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analysed in this study.

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